

JEE(Main + Advanced) : NURTURE COURSE (PHASE : TEPS)

PAPER-1

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16,
Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24,
Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127,
Xe = 131, Ba = 137, Ce = 140, Cu = 63.5, Ne = 20, K = 39, Mn = 55

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

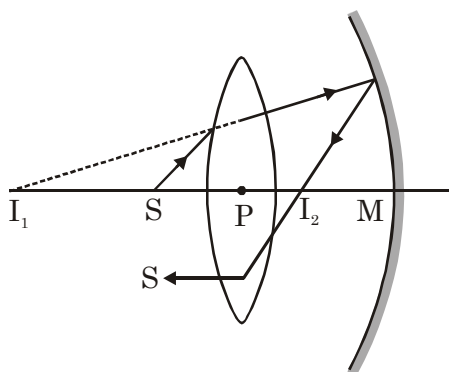
BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. A converging lens of focal length 15 cm and a converging mirror of focal length 20 cm are placed with their principal axes coinciding. A point source S is placed on the principal axis at a distance of 12 cm from the lens and in front of the lens. It is found that the final beam comes out parallel to the principal axis. Find the separation between the mirror and the lens.

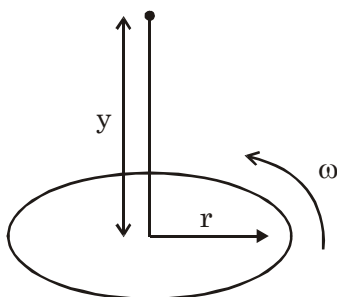


- (A) 60 cm (B) 40 cm (C) 25 cm (D) 20 cm
2. A circular ring of mass M and radius ' a ' is placed in a gravity free space. A small particle of mass m placed on the axis of the ring at a distance $\sqrt{3}a$ from the centre of the ring, is released from rest. The velocity with which the particles passes through the centre of the ring is:

- (A) $\sqrt{\frac{GM}{a}}$ (B) $\sqrt{\frac{GM}{2a} \left(\frac{m}{M} \right)}$ (C) $\sqrt{\frac{GM}{2a} \left(\frac{M}{m} \right)}$ (D) $\sqrt{\frac{GM}{a} \left(\frac{M}{M+m} \right)}$

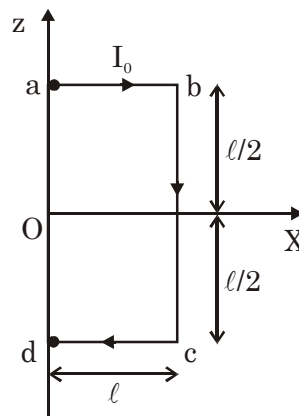
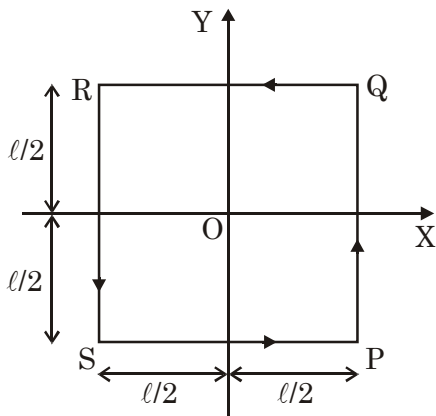
Space for Rough Work

3. Calculate the magnetic field at distance y from the centre of the axis of a disc of radius r and uniform surface charge density σ . if the disc spins with angular velocity ω ?



- (A) $\frac{\mu_0 \sigma \omega}{3} \left(\frac{r^2 - 2y^2}{\sqrt{r^2 - y^2}} + 2y \right)$ (B) $\frac{\mu_0 \sigma \omega}{2} \left(\frac{r^2 + y^2}{\sqrt{r^2 + y^2}} \right)$
- (C) $\frac{\mu_0 \sigma \omega}{2} \left(\frac{r^2 + 2y^2}{\sqrt{r^2 + y^2}} - 2y \right)$ (D) $\frac{2\mu_0 \sigma \omega}{3} \left(\frac{r^2 + 2y^2}{\sqrt{r^2 + y^2}} - 2y \right)$

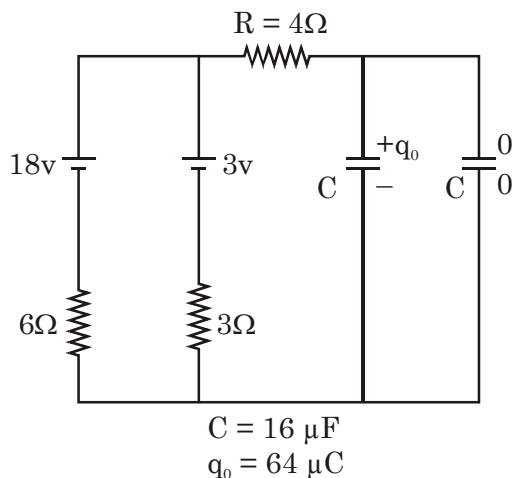
4. Current I_0 is flowing through a bent wire $a \rightarrow b \rightarrow c \rightarrow d$ in z - x plane as shown in figure, then $\oint \vec{B} \cdot d\vec{\ell}$ over the loop PQRS lying in the X - Y plane as shown in figure, due to the bent wire abcd :-



- (A) $\frac{\mu_0 I_0}{8}$ (B) $\frac{\mu_0 I_0}{7}$ (C) $\frac{\mu_0 I_0}{5}$ (D) $\frac{\mu_0 I_0}{3}$

Space for Rough Work

5. The circuit is connected at $t = 0$, then heat loss in $R = 4\Omega$ resistance, till the capacitor gets fully charged, is :



- (A) $362.67\ \mu\text{J}$ (B) $384\ \mu\text{J}$ (C) $426.67\ \mu\text{J}$ (D) $576\ \mu\text{J}$
6. A simple harmonic plane progressive wave is travelling along the +ve x-axis having amplitude A and $\omega = \pi\ \text{rad/s}$. At $t = 1\ \text{sec}$, particle at $x = 2\ \text{m}$ and $x = 3\ \text{m}$ are moving in downward and upward direction respectively at the position $y = +\frac{A}{2}$. The time when maximum power transfer will takes

place at $x = \frac{11}{4}\ \text{m}$, is

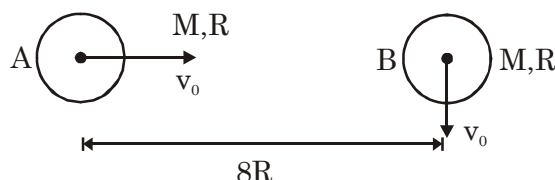
- (A) $\frac{5}{4}\ \text{sec}$ (B) $\frac{5}{3}\ \text{sec}$ (C) $\frac{7}{4}\ \text{sec}$ (D) $\frac{7}{5}\ \text{sec}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. Two spheres are moving under mutual gravitational force as shown. Velocity v_0 is such that spheres are at minimum separation when they touch each other. Then:



(A) $v_0 = \sqrt{\frac{GM}{4R}}$

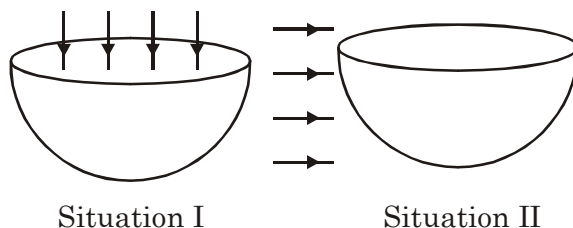
(B) $v_0 = \sqrt{\frac{15 GM}{56 R}}$

(C) Speed of sphere A at the time of minimum separation is $\sqrt{\frac{3 GM}{7 R}}$

(D) Speed of sphere B at the time of minimum separation is $\sqrt{\frac{6 GM}{7 R}}$

Space for Rough Work

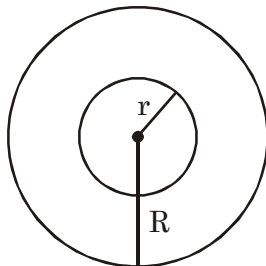
8. Two friends start cycling around a circular track (radius R) in the same sense starting from the same point with angular speed ω and 2ω . The slower one is continuously singing with his vocal cord frequency f_0 ($R\omega = 1\%$ of v_{sound}). Then : (regarding the frequency of song heard by the faster one)
- (A) Observed minimum frequency = $\frac{98}{99}f_0$.
- (B) Observed minimum frequency = $\frac{97}{99}f_0$.
- (C) Observed maximum frequency = $\frac{102}{101}f_0$
- (D) The spectrum (of frequencies) repeats itself with a time period of $\frac{\pi}{\omega}$.
9. In the figure a hemispherical bowl of radius R is shown. Electric field of intensity E is present perpendicular to the circular cross section of the hemisphere.



- (A) The magnitude of electric flux through the hemispherical surface in situation I will be $E\pi R^2$.
- (B) The magnitude of electric flux through the hemispherical surface in situation I will be $2E\pi R^2$.
- (C) The magnitude of electric flux through the hemispherical surface in situation II will be $\frac{E\pi R^2}{2}$.
- (D) The magnitude of electric flux through the hemispherical surface in situation II will be zero.

Space for Rough Work

10. A soap bubble of radius r is formed inside another soap bubble of radius $R(>r)$. The atmosphere pressure is P_0 and surface tension of the soap solution is T . Now the bigger bubble bursts. Assume that the excess pressure inside a bubble is small compared to P_0 . For the smaller bubble choose the **CORRECT** options:



(A) The bubble shrinks

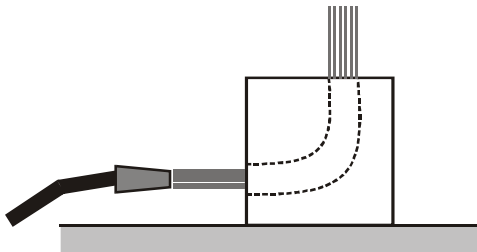
(B) The bubble expands

(C) Change in its radius is $\frac{4Tr}{3P_0R}$

(D) Change in its radius is $\frac{3Tr}{4P_0R}$

Space for Rough Work

11. A block of mass M , at rest on a frictionless surface, contains a curved channel. The channel is designed so that when a horizontal jet of fluid is aimed at it, the fluid is redirected 90 and shoots straight up and out of the top of the block. The flow of the jet is Q kg/s and has a velocity v m/s. Assuming no gravity and ignoring any viscous effects. [Density of fluid is ρ]



- (A) Speed of the block as a function of time is $v_b = v \left(1 - e^{-\frac{\rho Q t}{M}} \right)$
- (B) Speed of the block as a function of time is $v_b = v \left(\frac{Q t}{Q t + M} \right)$
- (C) Power delivered by the fluid to the block at $t = 0$ is zero
- (D) Power delivered by the fluid to the block at $t = 0$ is Qv
12. Velocity and acceleration vector of a charged particle moving in a magnetic field at some instant are $\vec{v} = 3\hat{i} + 4\hat{j}$ and $\vec{a} = 2\hat{i} + x\hat{j}$. Select the correct alternative (s)
- (A) $x = -1.5$
- (B) $x = 3$
- (C) Magnetic field is along z -direction
- (D) Kinetic energy of the particle is constant

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

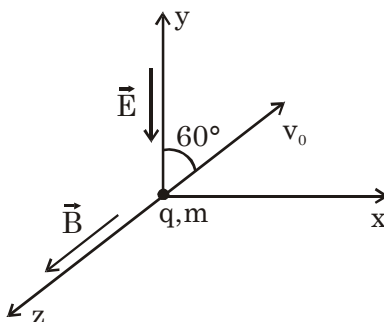
- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

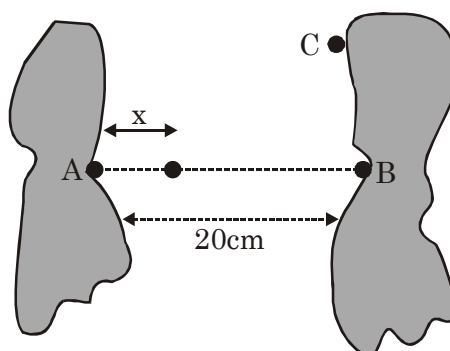
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A positively charged particle is given an initial velocity v_0 in XY plane as shown in diagram. Initially the particle was at origin and the region contains uniform, mutually perpendicular electric and magnetic field. Let $\vec{E} = -E\hat{j}$ and $\vec{B} = B\hat{k}$. The minimum magnitude of the magnetic field so that the charge particle crosses the origin again during its subsequent motion is $B_{\min} = \frac{n\pi E}{v_0}$, find the integer value of n.



Space for Rough Work

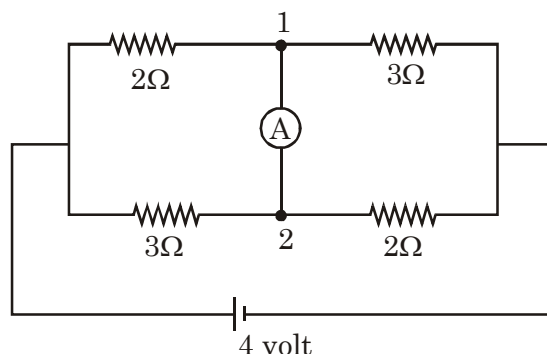
2. Two identical parallel plate capacitors are connected in parallel and charged to a voltage $v_0 = 12 \text{ V}$. When you disconnect them from the voltage source and the distance between the plates of one of the capacitors is reduced to one third of the original distance, what will be the new voltage on the capacitors (in volts) ?
3. Two metallic bodies separated by a distance 20cm are given equal and opposite charges of magnitude $0.88\mu\text{C}$. The component of electric field along the line AB, between the plates, varies as $E_x = 3x^2 + 0.4 \text{ N/C}$, where x (in meters) is the distance from one body towards the other as shown. The capacitance of the system is $n \times 10\mu\text{F}$. Find n



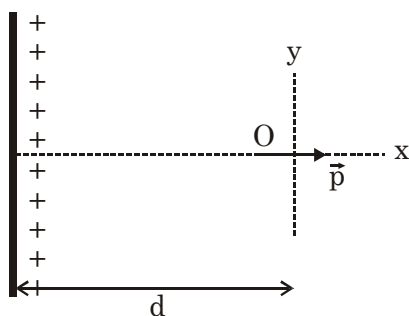
4. A metallic spherical shell has mass m , radius R and Bulk modulus of elasticity B . The change in radius of the shell due to gravitational pressure is $\frac{Gm^2}{n\pi BR^3}$. Find the value of n .

Space for Rough Work

5. In the circuit shown, the reading of the ammeter is $\frac{p}{q}$ amp. What is the value of $p + q$?
(Take : p and q to be smallest integers.)



6. An infinite non-conducting plane with uniform charge density σ is kept parallel to yz plane and at a distance ' d ' from a dipole $\vec{p}$ which itself is located at the origin. An equipotential surface for this system is spherical, centred at origin, having radius $R(<d)$. Given that $R = \left(\frac{p}{n\pi\sigma}\right)^{1/3}$; find the integer n .



Space for Rough Work

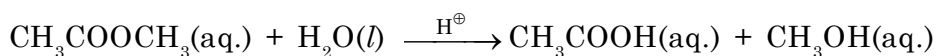
PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks : 18)**

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- Solution showing positive deviation from Raoult's law is-
 (A) Ethanol and acetone (B) Ethanol and methanol
 (C) HNO_3 & H_2O (D) Bromoethane and chloroethane
- Methyl acetate was hydrolyzed in HCl aqueous solution at 298K, samples of reaction mixture were withdrawn at different time intervals and titrated with 0.12N NaOH sample whose volumes were obtained as :

Time(mins)	0	25	∞
Volume of NaOH(mL)	18	30	42

Half- life for the reaction below is-



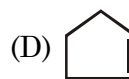
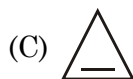
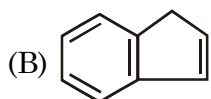
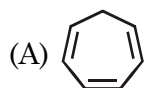
- (A) 12.5 minutes (B) 25 minutes (C) 37.5 minutes (D) 50 minutes
- Which of the following hydrolysis is **NOT** a type of SN^2 reaction?
 (A) $\text{SiCl}_4 + \text{H}_2\text{O} \longrightarrow$ (B) $\text{BCl}_3 + \text{H}_2\text{O} \longrightarrow$
 (C) $\text{CCl}_4 + \text{H}_2\text{O} \longrightarrow$ (D) $\text{XeF}_6 + \text{H}_2\text{O} \longrightarrow$

Space for Rough Work

4. Select the **INCORRECT** order?

- (A) $\text{LiOH} < \text{NaOH} < \text{KOH} < \text{RbOH}$ (thermal stability)
 (B) $\text{LiCl} > \text{NaCl} > \text{KCl} > \text{RbCl}$ (solubility)
 (C) $\text{D}_2\text{O} > \text{H}_2\text{O}$ (boiling point)
 (D) $\text{AgCl} < \text{AgBr} < \text{AgI}$ (colour intensity)

5. Which of following shows maximum acidic strength :



6. Select most acidic compound :

- (A) Ethane (B) Ethene (C) Ethyne (D) Acetone

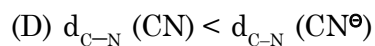
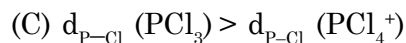
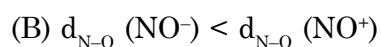
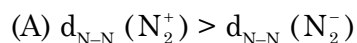
Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

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 - **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
7. In context to Maxwell's distribution of speed among gaseous molecules choose the correct statement(s)
- (A) With increase in temperature, the speed of moving molecules decreases
(B) With increase in temperature, average speed of gaseous molecules increases with increase in fraction of molecules having high speed
(C) Most probable speed of gas decreases with decrease in temperature
(D) With increase in temperature of gas total area of molecular speed distribution curve increases
8. Select the correct statement(s)-
- (A) In FCC, each atom is surrounded by six octahedral voids.
(B) In FCC , each atom is surrounded by eight tetrahedral voids.
(C) In simple cubic, each atom is surrounded by eight cubical voids
(D) In hcp, each atom is surrounded by four tetrahedral voids

Space for Rough Work

9. Select the **INCORRECT** bond length order(s).



10. Select the **INCORRECT** match(s)?

Salt/silicate	Total number of shared oxygen in anionic part of Salt/silicate
(A) $\text{Be}_3\text{Al}_2(\text{Si}_6\text{O}_{18})$	6
(B) $\text{Zn}_3(\text{Si}_2\text{O}_7)$	2
(C) Sodium tetrapolyphosphate	4
(D) Sodium trimetaborate	3

Space for Rough Work

11. Which of the following is/are incorrect IUPAC name ?

- (A) 4-Chloro-3-methyl cyclopentanol-1
- (B) 1-Amino-3-bromo hexan-1-one
- (C) 4-Chloro-3-methyl cyclohexane carboxylic acid
- (D) 3-Bromo-1-methyl hexan-1-ol

12. Which of the following order are correct ?

- (A) $\text{Ph}_3\text{C}^\ominus > \text{Ph}_2\text{CH}^\ominus > \text{PhCH}_2^\ominus$ (carbanion stability)
- (B) $-\text{NMe}_3^+ > -\text{NH}_3^+ > -\text{NH}_2 > -\text{OH}$ (order of $-I$ effect)
- (C) $\text{CH}_3 - \text{CH}_2^+ > \text{CH}_2 = \text{CH}^+ > \text{HC} \equiv \text{C}^+$ (carbocation stability)
- (D) $-\text{CMe}_3 > -\text{CHMe}_2 > -\text{CH}_3 > -\text{O}^-$ (order of $+I$ effect)

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

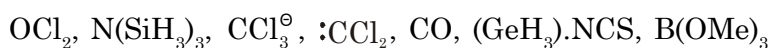
-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

- The density of steam at 227°C and 10^5 Pa is 0.6 kg/m³. Calculate compressibility factor for the steam. (Use : R = 8 J/K-mole).
- An aqueous solution of N₂ gas obeys Henry's law at very low concentration of gas in water ($n_{N_2} \ll n_{\text{water}}$). How many moles of N₂ gas will dissolve in 18 kg. water at 27°C and 1.6×10^7 Pa ? K_H (Henry's law constant) for N₂ gas in water at 27°C = 80 K bar.
- Among the following species in which highest energy electron in their ground state electronic configuration is present in antibonding molecular orbital, find sum of bond order of all such species.
O₂²⁺, N₂⁺, B₂, C₂²⁻, O₂⁻, NO⁻, CO, F₂

Space for Rough Work

4. Consider the following species.



Find the value of x.y if

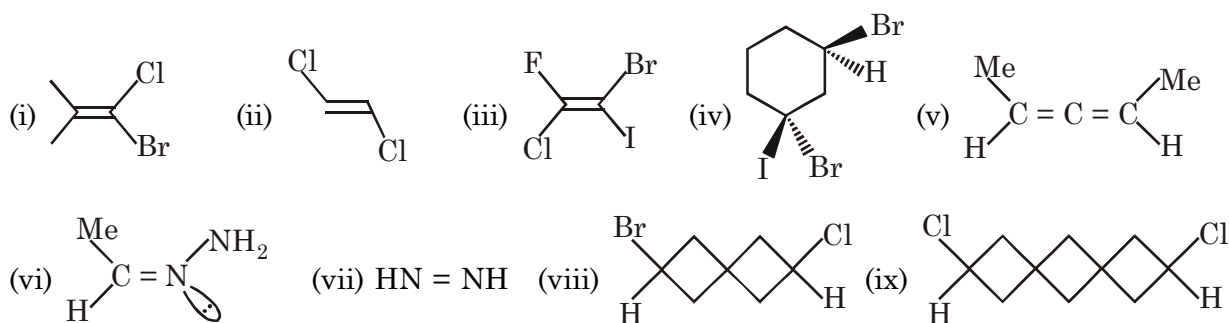
x = number of species in which $2p_\pi-3d_\pi$ back bond present

y = number of species in which direction of lone pair donation is from 2nd period element to 3rd period element

5. How many compounds are named as acid but does not have carboxylic acid group ?

- (a) Malonic acid (b) Picric acid (c) Squaric acid (d) Carbolic acid
(e) Crotonic acid (f) Cinnammic acid

6. How many of the following can show geometrical isomerism ?



Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 18)

- This section contains **SIX** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If $L = \lim_{x \rightarrow 2^+} \frac{[x-2]}{\operatorname{sgn}(x-2)}$ & $M = \lim_{x \rightarrow 2^-} \frac{[x-2]}{\operatorname{sgn}(x-2)}$, where $[.]$ denotes greatest integer function & $\operatorname{sgn}(x)$ denotes signum function), then $L + M$ is equal to -
 (A) 1 (B) 0 (C) -1 (D) 2
2. If $f(x) = \begin{cases} a \cot^2 x \cdot \ln \cos x, & x < 0 \\ 1, & x = 0 \\ (\sec x)^{b \cot x}, & x > 0 \end{cases}$ is continuous at $x = 0$, then -
 (A) $a = -2, b \in \mathbb{R}$ (B) $a = -2, b = 0$ (C) $a = -2, b = e$ (D) $a = -2, b = \frac{1}{e}$
3. Consider $f(x) = x^2 - 5x + 4$, then number of solutions of $|f(|x|)| = 2$ is equal to -
 (A) 2 (B) 4 (C) 6 (D) 8

Space for Rough Work

4. The number of arrangements of the letters of the word 'COMBINATIONS' in which three vowels occurs in odd positions and two different vowels occurs in even positions, is-
- (A) $7200 \times 7!$ (B) $2700 \times 7!$
 (C) $4500 \times 7!$ (D) $3600 \times 7!$
5. Let $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and $\frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$ are confocal ($a > A$ & $a > b$) having the foci at S_1 & S_2 . Let P be their point of intersection, then S_1P & S_2P are the roots of quadratic equation -
- (A) $x^2 + 2ax + (a^2 - A^2) = 0$ (B) $x^2 - 2Ax + (a^2 + A^2) = 0$
 (C) $x^2 - 2Ax + (a^2 - A^2) = 0$ (D) $x^2 - 2ax + (a^2 - A^2) = 0$
6. If the domain of $f(x)$ is $[-1, 2]$ then the domain of the function $f([x] - x^2 + 4)$ is (where $[.]$ denotes greatest integer function)-
- (A) $[-1, \sqrt{7}]$ (B) $[-\sqrt{3}, -1] \cup [\sqrt{3}, \sqrt{7}]$
 (C) $[-\sqrt{3}, \sqrt{7}] - (-1, 2)$ (D) $[\sqrt{13}, \sqrt{15}]$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

7. If $f(x)$, $g(x)$ are continuous & $h(x)$ is discontinuous at $x = a$, then -
- (A) $f(x) + g(x) - h(x)$ may or may not be continuous at $x = a$
- (B) $f(x) g(x) h(x)$ may or may not be continuous at $x = a$
- (C) $f(x) + g(x)$ is always continuous at $x = a$
- (D) $\frac{f(x)+g(x)}{h(x)}$ may or may not be continuous at $x = a$
8. Let $f(x) = \frac{2x+1}{x}$, $x \neq 0$, where $g(x) = f(f(x)) - \frac{1}{f(x)}$, $x \neq 0, -\frac{1}{2}$, then identify the correct statement(s) -
- (A) $g(x)$ is an odd function
- (B) $g(x)$ is an even function
- (C) $g(x)$ is neither even nor odd
- (D) $g(x)$ is a bounded function

Space for Rough Work

9. If $\sin^{-1}\left(\frac{\sqrt{x}}{2}\right) + \sin^{-1}\left(\sqrt{1-\frac{x}{4}}\right) + \tan^{-1}(y) = \frac{2\pi}{3}$, then

(A) Maximum value of $x^2 + y^2$ is $\frac{49}{3}$

(B) Maximum value of $x^2 + y^2$ is 4

(C) Minimum value of $x^2 + y^2$ is $\frac{1}{3}$

(D) Minimum value of $x^2 + y^2$ is 3

10. Consider the parabola $y^2 = 12x$. Then

(A) line $y = 3x + 1$ is a tangent to the parabola.

(B) line $y = 2x - 36$ is a normal to the parabola.

(C) line $2x + y = 6$ is a chord of contact for some point on directrix of the parabola.

(D) line $x - 2y + 12 = 0$ subtends a right angle at the vertex of the parabola.

Space for Rough Work

11. If $\frac{x}{ma} + \frac{y}{nb} = 1$ touches the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (n,m, a & b are positive real numbers), then -

(A) $m^2 = \frac{n^2}{n^2 - 1}$

(B) $n^2 = \frac{m^2}{m^2 - 1}$

(C) $m^2 = \frac{n^2}{n^2 + 1}$

(D) $n^2 = \frac{m^2}{m^2 + 1}$

12. If $\lim_{x \rightarrow 0} (a \sin bx^2 - 4 \cos^2 x + 5)^{\frac{4}{5x^2}} = e^2$ where a,b $\in \mathbb{R}$ then the value of a & b can be -

(A) $-3, \frac{1}{2}$

(B) $3, -\frac{1}{2}$

(C) $6, -\frac{1}{4}$

(D) $1, -\frac{3}{2}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

		+			
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	●	7	7
8	8	8	8	8	8
9	9	9	9	9	9

		-			
●	●	0	0	●	●
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	●	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Consider $f(x) = \sin^{-1}[2x] + \cos^{-1}([x] - 1)$ (where $[.]$ denotes greatest integer function.) If domain of $f(x)$ is $[a, b]$ & the range of $f(x)$ is $\{c, d\}$ then $a + b + \frac{2d}{c}$ is equal to (where $c < d$)
2. Let $f(x) = \sec(\operatorname{cosec}^{-1}x)$, then $\lim_{x \rightarrow 2} \frac{\sqrt{3} \left(\frac{x}{f(x)} - \sqrt{3} \right)}{(x-2)}$ is equal to
3. If two circles $x^2 + y^2 + 2n_1x + 2y + \frac{1}{2} = 0$ and $x^2 + y^2 + n_2x + n_2y + n_1 = \frac{1}{2}$, intersect each other orthogonally where $n_1, n_2 \in \mathbb{I}$, then number of possible of ordered pairs (n_1, n_2) is

Space for Rough Work

4. If the value of ${}^{40}C_4 - {}^4C_1 \cdot {}^{30}C_4 + {}^4C_2 \cdot {}^{20}C_4 - {}^4C_3 \cdot {}^{10}C_4$ is (a^b) where b is an even prime number, then sum of the digits of 'a' is
5. $P(x)$ be a polynomial satisfying $P(x) - 2P'(x) = 3x^3 - 27x^2 + 38x + 1$.

If function $f(x) = \begin{cases} \frac{P''(x)+18}{6} & x \neq \frac{\pi}{2} \\ \sin^{-1}(ab) + \cos^{-1}(a+b-3ab), & x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$, then $(a + b)$ is

equal to

6. If $\tan^{-1} \frac{2}{1^2} + \tan^{-1} \frac{2}{2^2} + \tan^{-1} \frac{2}{3^2} + \tan^{-1} \frac{2}{4^2} + \dots \dots \dots \tan^{-1} \frac{2}{17^2} = \pi - \tan^{-1} \frac{p}{q}$, where p & q are co-prime, then $(p - q)$ is

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	Single correct option	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	–1 In all other cases	18
I(ii)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	–2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator